Technical Evaluation Report

Vector Database Benchmarking Study

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Abstract

This report evaluates three vector database backends (Redis, ChromaDB, Weaviate) across five dimensions: indexing throughput, query latency, recall accuracy, memory footprint, and operational complexity. Benchmarks were conducted on a corpus of 500,000 document chunks derived from publicly available research papers. Results indicate that Redis Stack achieves the best latency profile while ChromaDB offers the simplest operational model for development environments.

1. Introduction

Retrieval-Augmented Generation (RAG) pipelines depend critically on the performance of the underlying vector store. As corpus sizes grow into the millions of chunks, the choice of vector database has significant implications for both system latency and infrastructure cost. This study provides empirical data to guide that decision in production deployments.

2. Methodology

All benchmarks were run on identical hardware: 16-core AMD EPYC, 64GB RAM, NVMe SSD. Each backend was configured with default settings except for index parameters which were tuned to the ef_construction=200, M=16 HNSW baseline. Queries were drawn from a held-out set of 10,000 natural language questions. Recall@10 was computed against a brute-force exact search ground truth.

3. Results

Table 1 summarises the benchmark results. Latency is reported as p50/p99 in milliseconds. Throughput is vectors indexed per second. Recall is Recall@10.

Backend	Index Throughput	p50 Latency	p99 Latency	Recall@10	RAM (GB)
Redis Stack	48,200 v/s	3.1 ms	9.4 ms	0.962	8.2
ChromaDB	22,100 v/s	7.8 ms	24.1 ms	0.971	4.1
Weaviate	39,800 v/s	4.2 ms	12.8 ms	0.958	11.4

4. Conclusion

Redis Stack is recommended for latency-sensitive production workloads. ChromaDB is suitable for development and small-scale deployments due to its zero-infrastructure footprint. Weaviate offers a middle ground with strong recall and reasonable operational overhead. Future work will extend these benchmarks to sparse-dense hybrid retrieval configurations.

References

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